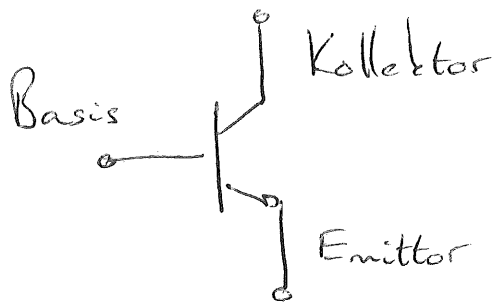
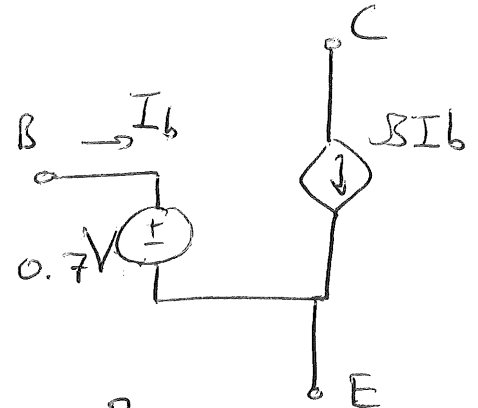


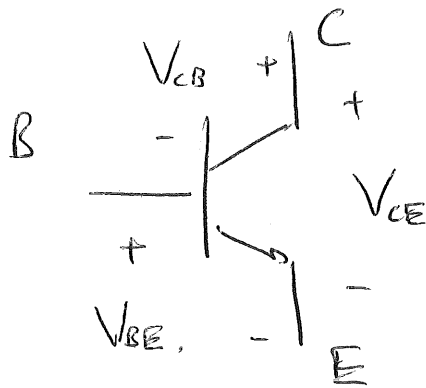
Transistors



≡
Ekwivalente
kringbaan



→ Current Controlled
Current Source!

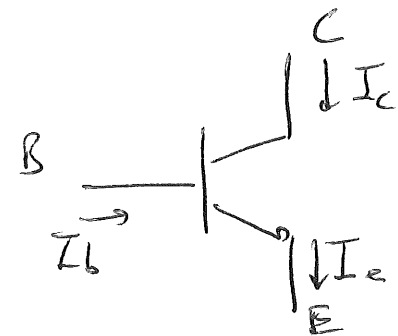


* $V_{BE} = 0.7V$

(V_{BE} is always 0.7V!)

* KVL:

$+V_{BE} - V_{CE} + V_{CB} = 0.$



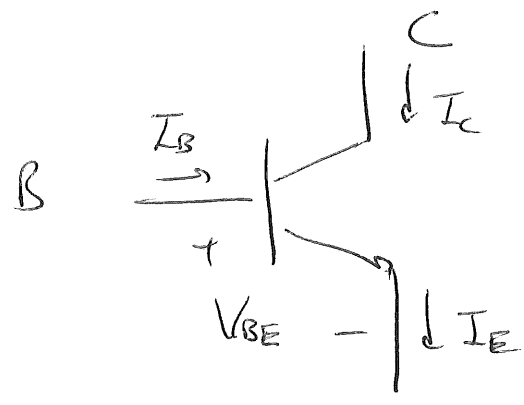
* $I_c = \beta I_b$

(β is a konstante)

* KCL:

$I_e = I_b + I_c$

* Om te onthou.



* $V_{BE} = 0.7V$

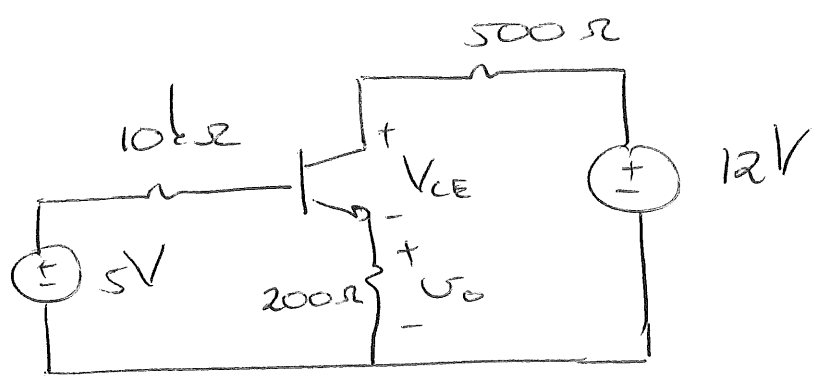
* $I_C = \beta I_B$

* $I_E = I_C + I_B$

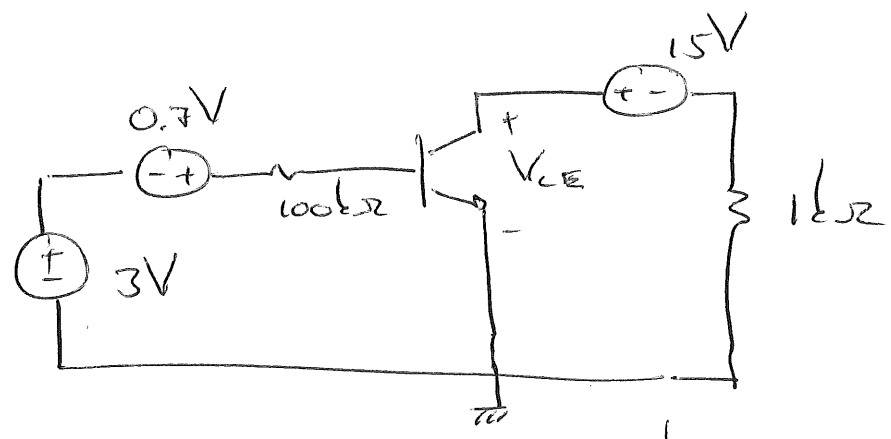
$(I_E = (\beta + 1) I_B)$

Voorbeeld:

- ① * $V_{BE} = 0.7V$
 * $\beta = 100$
 * $V_o = ?$
 * $V_{CE} = ?$



- ② * $V_{BE} = 0.7V$
 * $\beta = 100$
 * $I_B = ?$
 * $V_{CE} = ?$



- ③ * $V_{BE} = 0.7V$
 * $\beta = 0$
 * $V_o = ?$
 * $I_o = ?$

